

claude-windows / accd47af-95a3-46a7-9b51-b3eb5066a777

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\accd47af-95a3-46a7-9b51-b3eb5066a777\subagents\workflows\wf\_4fd1a4ea-6c0\agent-aa56b651c70bdad22.jsonl`
- SHA-256: `ed78e07d9d440d78aafd95699025f5f42ba5922c2fa63cf7a32f04f836f59d9e`
- Source modified: `2026-07-02T09:36:43+00:00`
- Imported at: `2026-07-05T16:48:24+00:00`
- project: `wf\_4fd1a4ea-6c0`
- session_id: `accd47af-95a3-46a7-9b51-b3eb5066a777`

Transcript

1. user (2026-07-02T09:30:45.756Z)

ADVERSARIAL VERIFIER. A reviewer of the P23 main.py->orchestration.py extraction claims:

TITLE: Verified OK ? all four requested seam checks pass against the current worktree state (162 passed, 1 skipped)

SEVERITY: nit

DETAIL: (1) Every patch of a moved name still intercepts its consumer: `_remote_input_size_bytes` is consumed only by retained routes in `main.py` (`backend/main.py:643, 872, 1017`) whose bare lookups hit `main`'s module dict where the re-export bound the name ? `monkeypatch` rebinds that dict, so interception is preserved; `orchestration` never calls it internally.

`_execute_compile/_execute_processing` patches (`tests/test_async_jobs.py:503,506`) intercept because `job_worker` resolves them at call time through the `backend.main` module object. (2) `backend/api/job_worker.py:114` (`_main._execute_compile`) and `:122` (`_main._execute_processing`) are covered by the re-export block at `backend/main.py:306-316`;

`test_worker_dispatches_compile_kind_to_execute_compile` passes. (3) Retained `main.py` call sites of moved names ? `_cloud_available` (`backend/main.py:473`), `_remote_input_size_bytes` (`:643,:872,:1017`), `_resolve_compile/_CompileError` (`:1170,:1177`), `_finalize_reingest/_report_config_for_input` (`:1618,:1628`) ? all resolve via the from-import binding in `main`'s globals, and patching `backend.main.<name>` rebinds exactly that dict, so route-level interception matches master. (4) Direct `m.<name>` callers work via the re-export:

`m._execute_processing` (`tests/test_api_cloud_dispatch.py:151-704`, `tests/test_async_jobs.py:173,207,226` ? including the positional-shim path relying on `isinstance(db, _main.ProcessRequest)`), `m._report_config_for_input` (`tests/test_api_endpoints.py:833,839`), and `'from backend.main import ProcessRequest'` (`tests/test_locale_analytics.py:12` ? `ProcessRequest` is retained, not moved). Object-attribute patches (`m.storage.*`, `m.registry.*`, `m.segmenter_registry.*`) mutate the shared imported objects and reach `orchestration` regardless. Empirical run in the worktree: `pytest over test_api_endpoints, test_async_jobs, test_api_cloud_dispatch, test_api_annotations, test_api_assets, test_job_worker_module_seam, test_job_worker_loop, test_main_module_budget, test_edge_auth, test_locale_analytics` = 162 passed, 1 skipped. Caveat: the worktree was actively changing during review (`orchestration.py` mtime 14:54:54, `main.py` 14:45:27); findings are pinned to that state ? re-verify at commit time.

EVIDENCE: `wt-docs/backend/api/job_worker.py:114,122`; `wt-docs/backend/main.py:306-316,473,643,872,1017,1170,1177,1618,1628`; test run: 162 passed, 1 skipped in 47.17s

REFACTOR UNDER REVIEW (uncommitted, in worktree `C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/accd47af-95a3-46a7-9b51-b3eb5066a777/scratchpad/wt-docs`, branch `refactor/p23-orchestration-split` off `origin/master`):

- NEW `backend/orchestration.py` (739 lines): 10 members moved VERBATIM from `backend/main.py`: `_finalize_reingest`, `_report_config_for_input`, `_ingest_remote_segments`, `_cloud_available`, `_maybe_dispatch_remote`, `_execute_processing`, `_remote_input_size_bytes`, `_CompileError`, `_resolve_compile`, `_execute_compile`.

- backend/main.py 2360 -> 1703 lines; re-exports all 10 names via "from backend.orchestration import (...)" (noqa F401);
 - 5 now-unused imports removed (csv, math, Tuple, StitchingConfig, localize_watermark).
- Enumerated intentional deltas (everything else is AST-identical, machine-verified):
 - (1) _maybe_dispatch_remote: "req: ProcessRequest" -> string forward-ref "ProcessRequest" (TYPE_CHECKING import).
 - (2) _execute_processing: inserted "import backend.main as _main" after docstring; global_config/db_instance/
 - ProcessRequest resolved via _main.<name> at call time (the job_worker seam convention).
 - (3) two "# type: ignore[...]" comments (AST-invisible).
- NEW tests/test_main_module_budget.py: line budget 1750, route ceiling 18, re-export identity check.
- Verified already: AST-identity (10 moved + 51 retained members), ruff+format clean, mypy green (142 files),
 - ratchet test passes. Full suite running separately.

Inspect with: git -C "C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/accd47af-95a3-46a7-9b51-b3eb5066a777/scratchpad/wt-docs" diff (and git -C "C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/accd47af-95a3-46a7-9b51-b3eb5066a777/scratchpad/wt-docs" status); the new files are untracked/modified in the worktree.

READ-ONLY: do not modify the worktree, do not checkout/commit.

Independently verify in the worktree (READ-ONLY). Default skeptical: real=true ONLY if you can reproduce the evidence AND the failure scenario is concrete (a test that would fail, a runtime path that breaks, an operational regression). A theoretical concern with no concrete consumer = real=false.

2. user (2026-07-02T09:30:55.074Z)

```
M backend/main.py
?? backend/orchestration.py
?? tests/test_main_module_budget.py
refactor/p23-orchestration-split
30736ab docs: make AUDIT_REPORT the umbrella code-audit doc; fold + archive the rest (#430)
```

3. user (2026-07-02T09:30:57.928Z)

```
__pycache__
annotations
api
assets
billing.py
compute
config
eval
features
flywheel.py
infrastructure
main.py
orchestration.py
pipeline
providers
reporting
results.py
sports
storage
tools
ui
utils
worker
1703 C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/accd47af-95a3-46a7-9b51-b3eb5066a777/scratchpad/wt-docs/backend/main.py
744 C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/accd47af-95a3-46a7-9b51-b3eb5066a777/scratchpad/wt-docs/backend/orchestration.py
2447 total
```

4. user (2026-07-02T09:31:16.696Z)

Found 8 files

```
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\accd47af-95a3-46a7-9b51-b3eb5066a777\scratchpad\wt-docs\tests\test_main_module_budget.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\accd47af-95a3-46a7-9b51-b3eb5066a777\scratchpad\wt-docs\tests\test_api_cloud_dispatch.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\accd47af-95a3-46a7-9b51-b3eb5066a777\scratchpad\wt-docs\tests\test_job_worker_module_seam.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\accd47af-95a3-46a7-9b51-b3eb5066a777\scratchpad\wt-docs\tests\test_base.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\accd47af-95a3-46a7-9b51-b3eb5066a777\scratchpad\wt-docs\tests\test_async_jobs.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\accd47af-95a3-46a7-9b51-b3eb5066a777\scratchpad\wt-docs\tests\test_api_endpoints.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\accd47af-95a3-46a7-9b51-b3eb5066a777\scratchpad\wt-docs\tests\test_api_annotations.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\accd47af-95a3-46a7-9b51-b3eb5066a777\scratchpad\wt-docs\tests\test_api_assets.py
```

5. user (2026-07-02T09:31:17.096Z)

The processing/compile ORCHESTRATION (hash?validate?segment?report pipeline, dispatch+ingest, compile resolution + worker handler) now lives in backend.orchestration (P23 god-module split; re-cut of #390). Re-exported here so the names stay addressable in backend.main ? `import backend.main as m; m.\_execute\_processing(...)` , the job worker call-time `\_main.\_execute\_compile(...)` seam, and monkeypatch.setattr(backend.main, all keep working unchanged.

```
from backend.orchestration import ( # noqa: F401 (re-exports: names stay addressable on
backend.main)
    _cloud_available,
    _CompileError,
    _execute_compile,
    _execute_processing,
    _finalize_reingest,
    _ingest_remote_segments,
    _maybe_dispatch_remote,
    _remote_input_size_bytes,
    _report_config_for_input,
    _resolve_compile,
)
```

Legacy thin wrapper for any remaining direct calls during transition.

```
import backend.main as _main

job_id = job["id"]
# Dispatch by kind: 'compile' (#329, the async reel stitch) vs 'process' (NULL/default ? the
# original /api/process pipeline). Both record their terminal state the same way; only
process
# jobs feed the M8 cost ledger + M11 flywheel (a compile has no detection usage/segments).
is_compile = job.get("kind") == "compile"
try:
    if is_compile:
        result = _main._execute_compile(
            config,
            db,
            job,
            progress=lambda stage: db.set_job_progress(job_id, stage),
```

```

    )
else:
    req = _main._job_to_request(job)
    result = _main._execute_processing(
        config,
        db,
        req,
        progress=lambda stage: db.set_job_progress(job_id, stage),
    )
    if result.get("video_id"):
        db.set_job_video_id(job_id, result["video_id"])
    db.mark_job_done(job_id, result)

```

6. user (2026-07-02T09:31:28.373Z)

```

C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\accd47af-95a3-46a7-9b51-b3eb5066a777\scratchpad\wt-docs\tests\test_api_assets.py:129:
monkeypatch.setattr(m, "_remote_input_size_bytes", lambda *a, **k: len(payload))
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\accd47af-95a3-46a7-9b51-b3eb5066a777\scratchpad\wt-docs\tests\test_api_annotations.py:91:
monkeypatch.setattr(m, "_remote_input_size_bytes", lambda *a, **k: len(payload))
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\accd47af-95a3-46a7-9b51-b3eb5066a777\scratchpad\wt-docs\tests\test_api_endpoints.py:815:
mp.setattr(m, "_remote_input_size_bytes", lambda *a, **k: len(payload))
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\accd47af-95a3-46a7-9b51-b3eb5066a777\scratchpad\wt-docs\tests\test_api_endpoints.py:851:
mp.setattr(m, "_remote_input_size_bytes", lambda *a, **k: 10 * 1024**3)
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\accd47af-95a3-46a7-9b51-b3eb5066a777\scratchpad\wt-docs\tests\test_api_endpoints.py:868:
mp.setattr(m, "_remote_input_size_bytes", lambda *a, **k: None)
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\accd47af-95a3-46a7-9b51-b3eb5066a777\scratchpad\wt-docs\tests\test_async_jobs.py:503:
mp.setattr(m, "_execute_compile", _fake_compile)

```

7. user (2026-07-02T09:31:37.114Z)

```

def _fake_compile(cfg, db, job, progress=None):
    seen["id"] = job["id"]
    return {
        "status": "success",
        "video_id": "vidX",
        "download_url": "/api/highlights/vidX/r.mp4",
    }

mp.setattr(m, "_execute_compile", _fake_compile)
mp.setattr(
    m,
    "_execute_processing",
    lambda *a, **k: pytest.fail("a compile job must not hit _execute_processing"),
)
db.create_compile_job("cj2", "vidX", "/v.mp4", [1], "r.mp4")
m._run_claimed_job(db.get_job("cj2"), m.global_config, m.db_instance)
assert seen["id"] == "cj2"
done = db.get_job("cj2")
assert done["status"] == "done"
assert done["result"]["download_url"] == "/api/highlights/vidX/r.mp4"

def test_create_job_persists_and_reconstructs_compute(tmp_path):
    """v8: the compute-picker choice must round-trip create_job ? get_job ? _job_to_request so
    the
    worker honours it across the async submit?reconstruct boundary. NULL ? server default (the
    pre-picker default-OFF behaviour). Regression guard for the dropped-across-boundary bug."""
    db = Database(str(tmp_path / "jobs.db"))

```

```

db.create_job("j-cloud", "gs://b/clip.mp4", compute="cloud")
db.create_job("j-default", "/v.mp4") # no compute ? NULL ? server default
assert db.get_job("j-cloud")["compute"] == "cloud"
assert db.get_job("j-default")["compute"] is None
assert m._job_to_request(db.get_job("j-cloud")).compute == "cloud"
assert m._job_to_request(db.get_job("j-default")).compute is None

```

--- crash recovery -----

```

def test_fail_stale_jobs_terminalizes_queued_and_running_only(env):
    client, db, tmp_path, mp = env
    db.create_job("q1", "/v.mp4")
    db.create_job("r1", "/v.mp4")
    db.mark_job_running("r1")
    db.create_job("d1", "/v.mp4")
    db.mark_job_done("d1", {"status": "success"})

```

8. user (2026-07-02T09:31:38.482Z)

```

307:     _cloud_available,
308:     _CompileError,
311:     _finalize_reingest,
312:     _ingest_remote_segments,
313:     _maybe_dispatch_remote,
314:     _remote_input_size_bytes,
315:     _report_config_for_input,
316:     _resolve_compile,
338:     # default (deployment.compute_target). Only honoured for these two values; see
    _maybe_dispatch_remote.
473:         "cloud_available": _cloud_available(cfg),
643:         need = _remote_input_size_bytes(input_ref, getattr(cfg, "storage", None))
872:         need = _remote_input_size_bytes(src_ref, getattr(cfg, "storage", None))
1017:         need = _remote_input_size_bytes(
1170:         video, _kept, out_name = _resolve_compile(
1177:     except _CompileError as e:
1618:         _finalize_reingest(db_instance, video_id, segments, play_wm, cand_wm)
1628:         config=_report_config_for_input(global_config, args.video_path),

```

9. user (2026-07-02T09:31:40.767Z)

```

4:Cloud Run dispatch + ingest path (`_maybe_dispatch_remote` / `_ingest_remote_segments`),
5:and the shared compile resolution + worker handler (`_resolve_compile` /
`_execute_compile`).
46:def _finalize_reingest(db, video_id: str, segments, play_wm: int, cand_wm: int) -> None:
59:def _report_config_for_input(config: Any, raw_input: str) -> Any:
87:def _ingest_remote_segments(
137:def _cloud_available(cfg) -> bool:
156:def _maybe_dispatch_remote(
216:     if not _cloud_available(cfg):
299:         _ingest_remote_segments(db, video_id, result.outputs_uri, storage_dict)
307:     # Only the NEW cloud rows (id > the pre-run watermark); _finalize_reingest then drops
the old ones.
311:     _finalize_reingest(db, video_id, segments, play_wm, cand_wm)
412:     # _maybe_dispatch_remote when it dispatches. None ? nothing ran (validation/config
fail).
498:         remote = _maybe_dispatch_remote(
535:         _finalize_reingest(db, video_id, segments, play_wm, cand_wm)
579:         config=_report_config_for_input(config, req.video_path),
594:def _remote_input_size_bytes(input_ref, storage_config) -> Optional[int]:
615:class _CompileError(Exception):
625:def _resolve_compile(
632:     ``_CompileError(404/400)``. Returns ``(video_row, kept_segments, safe_out_name)``.

```

```
635:         raise _CompileError(404, "Indexed video record not found.")
640:         raise _CompileError(400, str(e)) from e
645:         raise _CompileError(400, "No matching play segments found to stitch.")
666:         raise _CompileError(
689:             video, kept, out_name = _resolve_compile(
692:         except _CompileError as e:
```

10. user (2026-07-02T09:31:57.383Z)

```
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\accd47af-95a3-46a7-9b51-b3eb5066a777\scratchpad\wt-
docs\tests\test_job_worker_module_seam.py:41:     wake-timer THROUGH
`backend.main._arm_wake_timer`, so patching it on
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\accd47af-95a3-46a7-9b51-b3eb5066a777\scratchpad\wt-
docs\tests\test_job_worker_module_seam.py:65:     """"_arm_wake_timer` must submit the re-drain
through `backend.main._get_executor`
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\accd47af-95a3-46a7-9b51-b3eb5066a777\scratchpad\wt-
docs\tests\test_job_worker_module_seam.py:66:     and `backend.main._drain_due_jobs`, so a
stubbed executor + a real (non-leaking)
```

11. user (2026-07-02T09:31:59.192Z)

```
<persisted-output>
Output too large (31.4KB). Full output saved to: C:\Users\avidu\claude\projects\C--Users-avidu-
Projects-badminton-highlight-indexer\accd47af-95a3-46a7-9b51-b3eb5066a777\tool-
results\toolu_01PEiY4cPgVKCCaArYYCX1Ld.txt
```

```
Preview (first 2KB):
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\accd47af-95a3-46a7-9b51-b3eb5066a777\scratchpad\wt-docs\tests\test_api_assets.py:129:
monkeypatch.setattr(m, "_remote_input_size_bytes", lambda *a, **k: len(payload))
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\accd47af-95a3-46a7-9b51-b3eb5066a777\scratchpad\wt-
docs\tests\test_api_annotations.py:24:     monkeypatch.setattr(m, "_get_executor", lambda:
_InlineExecutor())
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\accd47af-95a3-46a7-9b51-b3eb5066a777\scratchpad\wt-
docs\tests\test_api_annotations.py:91:     monkeypatch.setattr(m, "_remote_input_size_bytes",
lambda *a, **k: len(payload))
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\accd47af-95a3-46a7-9b51-b3eb5066a777\scratchpad\wt-docs\tests\test_api_endpoints.py:32:
monkeypatch.setattr(m, "_get_executor", lambda: _InlineExecutor())
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\accd47af-95a3-46a7-9b51-b3eb5066a777\scratchpad\wt-docs\tests\test_api_endpoints.py:93:
mp.setattr(m, "global_config", _FakeConfig())
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\accd47af-95a3-46a7-9b51-b3eb5066a777\scratchpad\wt-docs\tests\test_api_endpoints.py:305:
mp.setattr(m, "global_config", cfg)
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\accd47af-95a3-46a7-9b51-b3eb5066a777\scratchpad\wt-docs\tests\test_api_endpoints.py:345:
mp.setattr(m, "global_config", cfg)
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\accd47af-95a3-46a7-9b51-b3eb5066a777\scratchpad\wt-docs\tests\test_api_endpoints.py:411:
mp.setattr(m, "VideoCompiler", _FakeCompiler)
...
</persisted-output>
```

12. user (2026-07-02T09:32:15.642Z)

```
Exit code 1
sed: -e expression #1, char 153: unterminated `s' command
```

13. user (2026-07-02T09:32:23.251Z)

```
test_api_annotations.py:24: monkeypatch.setattr(m, "_get_executor", lambda:
_InlineExecutor())
test_api_annotations.py:91: monkeypatch.setattr(m, "_remote_input_size_bytes", lambda *a,
**k: len(payload))
test_api_assets.py:129: monkeypatch.setattr(m, "_remote_input_size_bytes", lambda *a, **k:
len(payload))
test_api_cloud_dispatch.py:102: monkeypatch.setattr(m, "db_instance", db)
test_api_cloud_dispatch.py:103: monkeypatch.setattr(m, "db_instance", db)
test_api_cloud_dispatch.py:112: monkeypatch.setattr(m, "global_config", cfg)
test_api_cloud_dispatch.py:113: monkeypatch.setattr(m, "global_config", cfg)
test_api_cloud_dispatch.py:114: monkeypatch.setattr(m, "_get_executor", lambda:
_InlineExecutor())
test_api_cloud_dispatch.py:240: monkeypatch.setattr(m, "db_instance", db)
test_api_cloud_dispatch.py:241: monkeypatch.setattr(m, "db_instance", db)
test_api_cloud_dispatch.py:243: monkeypatch.setattr(m, "global_config", cfg)
test_api_cloud_dispatch.py:244: monkeypatch.setattr(m, "global_config", cfg)
test_api_cloud_dispatch.py:245: monkeypatch.setattr(m, "global_config", cfg)
test_api_cloud_dispatch.py:372: monkeypatch.setattr(m, "db_instance", db)
test_api_cloud_dispatch.py:381: monkeypatch.setattr(m, "global_config", cfg)
test_api_cloud_dispatch.py:382: monkeypatch.setattr(m, "_get_executor", lambda:
_InlineExecutor())
test_api_cloud_dispatch.py:414: monkeypatch.setattr(m, "db_instance", db)
test_api_cloud_dispatch.py:418: monkeypatch.setattr(m, "global_config", cfg)
test_api_cloud_dispatch.py:547: monkeypatch.setattr(m, "db_instance", db)
test_api_cloud_dispatch.py:556: monkeypatch.setattr(m, "global_config", cfg)
test_api_cloud_dispatch.py:557: monkeypatch.setattr(m, "_get_executor", lambda:
_InlineExecutor())
test_api_cloud_dispatch.py:637: monkeypatch.setattr(m, "db_instance", db)
test_api_cloud_dispatch.py:638: monkeypatch.setattr(m, "db_instance", db)
test_api_cloud_dispatch.py:639: monkeypatch.setattr(m, "global_config", AppConfig())
test_api_cloud_dispatch.py:658: monkeypatch.setattr(m, "db_instance", db)
test_api_cloud_dispatch.py:659: monkeypatch.setattr(m, "db_instance", db)
test_api_cloud_dispatch.py:694: monkeypatch.setattr(m, "db_instance", db)
test_api_cloud_dispatch.py:695: monkeypatch.setattr(m, "global_config", AppConfig())
test_api_endpoints.py:1016: monkeypatch.setattr(m, "load_app_config", lambda *a, **k: cfg)
test_api_endpoints.py:1017: monkeypatch.setattr(m, "Database", lambda *a, **k: db)
test_api_endpoints.py:1018: monkeypatch.setattr(m, "global_config", cfg)
test_api_endpoints.py:1020: monkeypatch.setattr(m, "db_instance", db)
test_api_endpoints.py:1066: monkeypatch.setattr(m, "load_app_config", lambda *a, **k: cfg)
test_api_endpoints.py:1067: monkeypatch.setattr(m, "Database", lambda *a, **k: db)
test_api_endpoints.py:1068: monkeypatch.setattr(m, "global_config", cfg)
test_api_endpoints.py:1070: monkeypatch.setattr(m, "db_instance", db)
test_api_endpoints.py:305: mp.setattr(m, "global_config", cfg)
test_api_endpoints.py:32: monkeypatch.setattr(m, "_get_executor", lambda: _InlineExecutor())
test_api_endpoints.py:345: mp.setattr(m, "global_config", cfg)
test_api_endpoints.py:411: mp.setattr(m, "VideoCompiler", _FakeCompiler)
test_api_endpoints.py:455: mp.setattr(m, "VideoCompiler", _CaptureCompiler)
test_api_endpoints.py:486: mp.setattr(m, "VideoCompiler", _fake_ok_compiler())
test_api_endpoints.py:531: mp.setattr(m, "VideoCompiler", _BoomCompiler)
test_api_endpoints.py:575: mp.setattr(m, "VideoCompiler", _FakeCompiler)
test_api_endpoints.py:782: mp.setattr(m, "emit_indexer_report", fake_emit)
test_api_endpoints.py:815: mp.setattr(m, "_remote_input_size_bytes", lambda *a, **k:
len(payload))
test_api_endpoints.py:817: mp.setattr(m, "compute_video_id", fail_hash)
test_api_endpoints.py:851: mp.setattr(m, "_remote_input_size_bytes", lambda *a, **k: 10 *
1024**3)
test_api_endpoints.py:868: mp.setattr(m, "_remote_input_size_bytes", lambda *a, **k: None)
test_api_endpoints.py:93: mp.setattr(m, "global_config", _FakeConfig())
test_async_jobs.py:40: monkeypatch.setattr(m, "_get_executor", lambda: _InlineExecutor())
test_async_jobs.py:503: mp.setattr(m, "_execute_compile", _fake_compile)
test_async_jobs.py:558: mp.setattr(m, "_get_executor", lambda: pool)
test_edge_auth.py:233: monkeypatch.setattr(m, "_get_executor", lambda: _Noop())
test_fusion_hybrid.py:213: with patch("backend.pipeline.segmenters.fusion_hybrid.WasbRunner")
```

```

as W, patch(
test_fusion_hybrid.py:217:    ) as T,
patch("backend.pipeline.segmenters.fusion_hybrid.TrackNetConfig") as TC:
test_gemini.py:134:@patch("backend.pipeline.segmenters.gemini.provider_registry")
test_gemini.py:135:@patch("backend.pipeline.segmenters.gemini.get_video_duration")
test_gemini.py:136:@patch("backend.pipeline.segmenters.gemini.extract_video_chunk")
test_gemini.py:153:@patch("backend.pipeline.segmenters.gemini.provider_registry")
test_gemini.py:154:@patch("backend.pipeline.segmenters.gemini.get_video_duration")
test_gemini.py:155:@patch("backend.pipeline.segmenters.gemini.extract_video_chunk")
test_gemini.py:181:@patch("backend.pipeline.segmenters.gemini.provider_registry")
test_gemini.py:182:@patch("backend.pipeline.segmenters.gemini.get_video_duration")
test_gemini.py:183:@patch("backend.pipeline.segmenters.gemini.extract_video_chunk")
test_gemini.py:21:@patch("backend.pipeline.segmenters.gemini.provider_registry")
test_gemini.py:214:@patch("backend.pipeline.segmenters.gemini.provider_registry")
test_gemini.py:215:@patch("backend.pipeline.segmenters.gemini.get_video_duration")
test_gemini.py:22:@patch("backend.pipeline.segmenters.gemini.get_video_duration")
test_gemini.py:261:    with patch("backend.pipeline.segmenters.gemini.sport_registry") as
mock_sr:
test_gemini.py:273:    ), patch("backend.pipeline.segmenters.gemini.provider_registry") as
mock_pr:
test_gemini.py:57:@patch("backend.pipeline.segmenters.gemini.provider_registry")
test_gemini.py:58:@patch("backend.pipeline.segmenters.gemini.get_video_duration")
test_gemini.py:59:@patch("backend.pipeline.segmenters.gemini.extract_video_chunk")
test_gemini_refine.py:58:    ), patch("backend.eval.gemini_refine.provider_registry") as reg:
test_gemini_refine.py:87:    ), patch("backend.eval.gemini_refine.provider_registry") as reg:
test_job_worker_loop.py:46:    monkeypatch.setattr(m, "_get_executor", lambda:
_InlineExecutor())
test_job_worker_loop.py:50:    monkeypatch.setattr(m, "_arm_wake_timer", lambda delay, *a:
armed.append(delay))
test_job_worker_module_seam.py:48:    monkeypatch.setattr(m, "db_instance", db)
test_job_worker_module_seam.py:51:    monkeypatch.setattr(m, "_arm_wake_timer", lambda delay,
*a: armed.append(delay))
test_job_worker_module_seam.py:7:THROUGH the `backend.main` module object, so that
`monkeypatch.setattr(m, ...)`
test_job_worker_module_seam.py:74:    monkeypatch.setattr(m, "_get_executor", lambda:
_RecordingExecutor())
test_person_detector.py:117:@patch("backend.pipeline.detectors.person_detector.cv2.VideoCapture")
)
test_person_detector.py:42:@patch("backend.pipeline.detectors.person_detector.cv2.VideoCapture")
test_person_detector.py:77:@patch("backend.pipeline.detectors.person_detector.cv2.VideoCapture")
test_provider_gemini.py:11:    with patch("backend.providers.gemini.genai") as mock_genai,
patch(
test_provider_gemini.py:113:    with patch("backend.providers.gemini.genai") as mock_genai:
test_provider_gemini.py:152:    with patch("backend.providers.gemini.genai"):
test_provider_gemini.py:171:    with patch("backend.providers.gemini.genai"):
test_provider_gemini.py:184:    with patch("backend.providers.gemini.genai"):
test_provider_gemini.py:193:    with patch("backend.providers.gemini.genai"):
test_provider_gemini.py:209:    with patch("backend.providers.gemini.genai"):
test_provider_gemini.py:38:    with patch("backend.providers.gemini.genai") as mock_genai:
test_provider_gemini.py:74:    with patch("backend.providers.gemini.genai") as mock_genai:
test_provider_gemini.py:94:    with patch("backend.providers.gemini.genai") as mock_genai:
test_reporting.py:29:    monkeypatch.setattr("backend.reporting.OBSREPORT_AVAILABLE", False)
test_reporting.py:51:    monkeypatch.setattr("backend.reporting.OBSREPORT_AVAILABLE", False)
test_reporting.py:61:    monkeypatch.setattr("backend.reporting.OBSREPORT_AVAILABLE", False)
test_storage.py:284:    monkeypatch.setattr("backend.storage.gcs.import_module", fake_import)
test_tracknet_hybrid.py:30:    fn =
patch("backend.pipeline.segmenters.tracknet_hybrid.TrackNetRunner")(fn)
test_tracknet_hybrid.py:35:    fn =
patch("backend.pipeline.segmenters.trajectory_hybrid.get_video_duration")(fn)
test_tracknet_hybrid.py:36:    fn =
patch("backend.pipeline.segmenters.trajectory_hybrid.extract_video_chunk")(fn)
test_tracknet_hybrid.py:37:    fn =
patch("backend.pipeline.segmenters.trajectory_hybrid.provider_registry")(fn)
test_yolo_hybrid.py:105:@patch("backend.pipeline.detectors.person_detector.cv2.VideoCapture")
test_yolo_hybrid.py:106:@patch("backend.pipeline.segmenters.yolo_hybrid.get_video_duration")

```



```

test_yolo_hybrid.py:107:@patch("backend.pipeline.segmenters.yolo_hybrid.extract_video_chunk")
test_yolo_hybrid.py:108:@patch("backend.pipeline.segmenters.yolo_hybrid.provider_registry")
test_yolo_hybrid.py:157:@patch("backend.pipeline.detectors.person_detector.cv2.VideoCapture")
test_yolo_hybrid.py:158:@patch("backend.pipeline.segmenters.yolo_hybrid.get_video_duration")
test_yolo_hybrid.py:159:@patch("backend.pipeline.segmenters.yolo_hybrid.extract_video_chunk")
test_yolo_hybrid.py:160:@patch("backend.pipeline.segmenters.yolo_hybrid.provider_registry")
test_yolo_hybrid.py:178:@patch("backend.pipeline.detectors.person_detector.cv2.VideoCapture")
test_yolo_hybrid.py:179:@patch("backend.pipeline.segmenters.yolo_hybrid.get_video_duration")
test_yolo_hybrid.py:180:@patch("backend.pipeline.segmenters.yolo_hybrid.extract_video_chunk")
test_yolo_hybrid.py:181:@patch("backend.pipeline.segmenters.yolo_hybrid.provider_registry")
test_yolo_hybrid.py:228:@patch("backend.pipeline.detectors.person_detector.cv2.VideoCapture")
test_yolo_hybrid.py:263:@patch("backend.pipeline.detectors.person_detector.cv2.VideoCapture")
test_yolo_hybrid.py:264:@patch("backend.pipeline.segmenters.yolo_hybrid.get_video_duration")
test_yolo_hybrid.py:265:@patch("backend.pipeline.segmenters.yolo_hybrid.extract_video_chunk")
test_yolo_hybrid.py:266:@patch("backend.pipeline.segmenters.yolo_hybrid.provider_registry")
test_yolo_hybrid.py:316:@patch("backend.pipeline.detectors.person_detector.cv2.VideoCapture")
test_yolo_hybrid.py:317:@patch("backend.pipeline.segmenters.yolo_hybrid.get_video_duration")
test_yolo_hybrid.py:318:@patch("backend.pipeline.segmenters.yolo_hybrid.extract_video_chunk")
test_yolo_hybrid.py:319:@patch("backend.pipeline.segmenters.yolo_hybrid.provider_registry")
test_yolo_hybrid.py:362:@patch("backend.pipeline.detectors.person_detector.cv2.VideoCapture")
test_yolo_hybrid.py:363:@patch("backend.pipeline.segmenters.yolo_hybrid.get_video_duration")
test_yolo_hybrid.py:364:@patch("backend.pipeline.segmenters.yolo_hybrid.extract_video_chunk")
test_yolo_hybrid.py:365:@patch("backend.pipeline.segmenters.yolo_hybrid.provider_registry")
test_yolo_hybrid.py:57:@patch("backend.pipeline.detectors.person_detector.cv2.VideoCapture")
test_yolo_hybrid.py:58:@patch("backend.pipeline.segmenters.yolo_hybrid.get_video_duration")
test_yolo_hybrid.py:59:@patch("backend.pipeline.segmenters.yolo_hybrid.extract_video_chunk")
test_yolo_hybrid.py:60:@patch("backend.pipeline.segmenters.yolo_hybrid.provider_registry")

```

14. user (2026-07-02T09:34:24.992Z)

```

16: ``ProcessRequest``, ``compute_video_id``, ``emit_indexer_report``, ``VideoCompiler``.
362:     if req is None and db is not None and isinstance(db, _main.ProcessRequest):
365:         elif req is None and isinstance(config, _main.ProcessRequest):
370:             config = _main.global_config
372:             db = _main.db_instance
386:             video_id = _main.compute_video_id(local_video)
571:             paths = _main.emit_indexer_report(
679: import backend.main as _main # call-time seam: VideoCompiler stays patchable on
backend.main
702:     compiler = _main.VideoCompiler(

```

15. user (2026-07-02T09:34:38.131Z)

```

61: from backend.pipeline.stitcher.compiler import VideoCompiler
64: from backend.reporting import emit_indexer_report
74: from backend.utils.hashing import compute_video_id # canonical file-identity hashing
359: # the whole-file MD5 (compute_video_id, D3) so it joins the pipeline by content hash.
822: MD5 (``compute_video_id``, D3) so it joins the pipeline by content hash, and appears in
891:     video_id = compute_video_id(local_working)
1437:     compiler = VideoCompiler(
1559:         video_id = compute_video_id(local_video)
1621:         paths = emit_indexer_report(

```

16. assistant (2026-07-02T09:34:54.032Z)

Line-number claims for main.py (473, 643, 872, 1017, 1170, 1177, 1618, 1628) all match exactly. Now verify job_worker line numbers and then run the test subset.

17. user (2026-07-02T09:34:56.972Z)

```

11: `backend.main` module object at call-time (`import backend.main as _main`), never as a
77:     import backend.main as _main
105:     import backend.main as _main
114:         result = _main._execute_compile(

```

```
122:         result = _main._execute_processing(
221:     import backend.main as _main
241:     import backend.main as _main
258:     import backend.main as _main
```

18. user (2026-07-02T09:35:03.303Z)

```
from backend.orchestration import ( # noqa: F401 (re-exports: names stay addressable on
backend.main)
    _cloud_available,
    _CompileError,
    True
```

19. assistant (2026-07-02T09:35:11.268Z)

Both static claims check out. Now reproduce the empirical test run with the exact subset the reviewer listed.

20. user (2026-07-02T09:36:03.636Z)

```
..... [100%]
===== warnings summary =====
tests/test_api_endpoints.py::test_process_nonlocal_uri_skips_exists_and_fetches
tests/test_api_endpoints.py::test_process_remote_input_reports_use_durable_output_dir
tests/test_api_endpoints.py::test_process_bucket_uri_yields_queryable_video_id[gs://bucket/video
s/x.mp4]
tests/test_api_endpoints.py::test_process_bucket_uri_yields_queryable_video_id[gs://bucket/vide
os/x.mp4]
tests/test_api_cloud_dispatch.py::test_full_api_process_to_query_via_cloud
tests/test_api_cloud_dispatch.py::test_local_override_survives_api_boundary
tests/test_api_cloud_dispatch.py::test_cloud_override_survives_api_boundary_on_local_server
tests/test_api_cloud_dispatch.py::test_full_api_process_job_result_carries_cloud_provenance
C:\Users\avidu\AppData\Local\Programs\Python\Python313\Lib\site-
packages\google\auth\_default.py:113: UserWarning: Your application has authenticated using end
user credentials from Google Cloud SDK without a quota project. You might receive a "quota
exceeded" or "API not enabled" error. See the following page for troubleshooting:
https://cloud.google.com/docs/authentication/adc-troubleshooting/user-creds.
    warnings.warn(_CLOUD_SDK_CREDENTIALS_WARNING)

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
162 passed, 1 skipped, 8 warnings in 45.76s
```

21. user (2026-07-02T09:36:21.947Z)

```
135:     Exercise the exact duck-typing patterns used in main.py (both the _execute_processing
181:     This allows the except blocks in main.py (_execute_processing:193-198 and the
```

22. user (2026-07-02T09:36:23.538Z)

```
328:class ProcessRequest(BaseModel):
```

23. assistant (2026-07-02T09:36:31.694Z)

All checks reproduced. The reviewer's "Verified OK" claim holds against the current worktree state.

24. user (2026-07-02T09:36:42.949Z)

Structured output provided successfully